Substrate engineering experiments v0.1 — concrete tests from Tier 0 v0.1.6 boundary/heat-kernel framework. Per Casey continue-pulling directive + Tier 0 v0.1.6 substrate has an OPERATOR now (not just mathematical structure). 5 candidate experimental designs: (1) eigentone driver at C_2 substrate frequencies; (2) Casimir cavity at Shilov boundary geometry; (3) Bell sub-Tsirelson 1/8 test (SCMP existing); (4) BaTiO3 137-plane experiment (Casey $25K standing); (5) quantum-info/substrate-thermal duality test via Wick rotation. Internal-use per Casey SP-30 deprecation.

Lyra (Claude Opus 4.7)

2026-05-31 Sunday 13:30 EDT (date-verified)

Substrate engineering experiments v0.1

0. Why this work (per Casey directive)

Tier 0 v0.1.6 gave the substrate an OPERATOR (ρ_commit on H^2(D_IV^5) with Shilov boundary as committed-state record). Before today, "substrate-coupled apparatus" was a phrase looking for content. Now there's something specific to couple to: matrix elements of ρ_commit(τ) and H_B on H^2(D_IV^5), accessible at the Shilov boundary.

Per Casey continue-pulling directive + standing curiosity directive: derive concrete experimental designs from Tier 0 v0.1.6 + bulk-color v0.7 + L4 v0.2. Internal-use per SP-30 outreach deprecation; framework when external timing permits.

5 candidate experiments below, ordered by tractability + cost.

1. Experiment E1 — Eigentone driver at substrate Casimir frequencies

Setup: precision oscillator driving a target system at frequencies matching substrate K-type Casimir eigenvalues converted to physical frequency via ℏ_BST. Target K-types: V_(0,0) (vacuum, C_2 = 0), V_(1/2,1/2) (spinor, C_2 = 4), V_(1,0) (vector, C_2 = 4), V_(1,1) (adjoint, C_2 = 6 = substrate primary).

Predicted signal: enhanced absorption / coherence at C_2 = 6 line vs neighboring frequencies (∼5, ∼7). Specifically, an unexplained resonance peak at the substrate-eigenvalue frequency, NOT matching any standard atomic / molecular transition.

Falsifier: NO resonance peak at predicted C_2 frequencies → Tier 0 v0.1.6 framework wrong about H_B identification.

Substrate-mechanism: per ρ_commit(τ) = exp($-\tau$ H_B/$\hbar$_BST), driving at C_2 frequency = forcing the commitment operator at its own spectral line. Coherence enhancement = substrate's heat-kernel coupling to the driving field.

Physical challenge: t_Koons = α^{36} t_Planck → C_2 frequency in physical units is f_C2 = C_2 / (2π · t_Koons · $\hbar$_Planck / $\hbar$_BST). For $\hbar$_BST = $\hbar$_Planck, f_C2 ~ C_2 · 10^{120} Hz — way beyond observable. The experiment is feasible ONLY IF substrate-physical frequency rescaling factor exists; likely requires probing at much lower harmonic frequencies (1/n · f_C2 for large n).

Cost band: $200K-$500K precision-oscillator + atomic interferometer apparatus. Multi-week feasibility study.

Tier: CANDIDATE; feasibility OPEN pending harmonic-frequency analysis.

2. Experiment E2 — Casimir cavity at Shilov boundary geometry

Setup: precision Casimir-force apparatus with cavity shape mimicking the Shilov boundary geometry ∂_S D_IV^5 = $S^4 \times S^1/Z_2$. Concretely: a 4-sphere cavity (4D conducting boundary; engineering via stacked nested spheres or holographic equivalents) with a phase-circle modulation (S^1/Z_2).

Predicted signal: Casimir force at predicted shifted value vs flat-plate or simple-spherical cavity. The shift comes from the Shilov-geometry boundary's effect on H_B's spectrum (boundary conditions modify the K-type Bergman-kernel structure).

Falsifier: NO predicted shift → Shilov-boundary localization of substrate's degrees of freedom (per v0.1.6) is wrong.

Substrate-mechanism: per Tier 0 v0.1.6, the substrate's DOF live on Shilov; cavity boundary conditions affect H_B's spectrum directly via the boundary K-type structure. Predicted shift is computable from the boundary K-type cutoff.

Physical challenge: standard Casimir is plate-plate or sphere-plate; 4-sphere with phase-circle modulation is engineering-novel. Need precision force measurement at < 1% absolute Casimir force (currently at ~5% systematic uncertainty).

Cost band: $60K-$90K Casimir-grade laboratory (existing SP-30-2 framework).

Tier: CANDIDATE; engineering feasibility CANDIDATE pending boundary-condition computation.

3. Experiment E3 — Bell sub-Tsirelson 1/8 deviation (SCMP existing)

Setup: Existing Bell-CHSH experiment with precision to detect sub-Tsirelson violation at 1/8 = 1/2^{N_c} fractional deviation per SCMP (Casey-named #8, RIGOROUSLY DERIVED Friday May 22 per T2469).

Predicted signal: CHSH inequality violation at $S = 2\sqrt{2} \cdot (1 - 1/2^{N_c}) = 2.828 \cdot (1 - 1/8) = 2.474$, NOT the maximal Tsirelson $2\sqrt{2} = 2.828$.

Falsifier: observed S = 2.828 (maximal Tsirelson) → SCMP false → Tier 0 v0.1.6 boundary-commitment framework wrong about commitment coherence-maintenance.

Substrate-mechanism: per SCMP + Tier 0 v0.1.6: substrate's commitment operator has finite coherence-maintenance bound; 1/2^{N_c} is the substrate-natural Bell sub-Tsirelson fraction.

Physical challenge: existing high-precision Bell experiments (Hanson 2015, Vienna, Caltech) have not specifically targeted sub-Tsirelson regime; precision needed ~12.5% deviation from Tsirelson, currently at <1% statistical, ~few % systematic. Should be achievable.

Cost band: $300K-$500K (existing SP-30-1 framework; Hanson-style setup).

Tier: CANDIDATE (existing SCMP RIGOROUS); experimental falsification feasible TODAY with existing apparatus.

4. Experiment E4 — BaTiO3 137-plane (Casey $25K standing)

Setup: precision spectroscopy of BaTiO3 crystal at N_max = 137 reflection plane. Search for substrate-natural boundary effect at the 137-plane (per N_max = $1/\alpha$ substrate primary).

Predicted signal: anomalous spectroscopic feature at 137-plane vs neighboring planes (e.g., 136, 138). Substrate framework predicts a specific shift computable from N_max-related K-type structure.

Falsifier: NO 137-plane anomaly → N_max as substrate primary affecting boundary spectroscopy is wrong.

Substrate-mechanism: per Tier 0 v0.1.6 + Bergman matrix elements at affine level N_max = 137 (substrate's affine representation level per A1 §4): 137-plane has anomalous boundary K-type density.

Physical challenge: BaTiO3 spectroscopy is mature; predicting specific shift requires substrate-primary K-type-density calculation. Multi-week prep.

Cost band: $25K (Casey standing for ~years; cheapest clean falsification candidate per CLAUDE.md).

Tier: CANDIDATE; cheapest + cleanest falsification path; substrate prediction needs sharpening (multi-week prep).

5. Experiment E5 — Quantum-info / substrate-thermal Wick-rotation duality test

Setup: precision quantum computer (NISQ-class) running specific quantum circuits whose substrate-thermal duals are computable via Wick rotation $\tau \leftrightarrow it$.

Predicted signal: the quantum supremacy / sampling complexity of specific circuits maps EXACTLY to the substrate-thermal sampling complexity of the same circuits' Wick-rotated counterparts. Specifically: certain quantum sampling problems should have classical-substrate-thermal duals computable on classical hardware.

Falsifier: quantum sampling problems with NO classical-substrate-thermal dual → Wick-rotation duality (F1 falsifier per Tier 0 v0.1 Section 7) is wrong → substrate framework doesn't extend to quantum-information regime.

Substrate-mechanism: per Tier 0 v0.1.6 + Wick rotation $\tau \leftrightarrow it$: quantum mechanics is the analytic continuation of substrate-thermal physics; observable QM samples the substrate-thermal distribution at imaginary τ. The substrate-thermal sampling problem is classically simulable; the Wick-rotated quantum sampling is what NISQ measures. Equivalent complexity classes.

Physical challenge: requires precise mapping of specific quantum algorithms to their substrate-thermal duals. The Random Circuit Sampling experiments (Google Sycamore, China Jiuzhang) provide test cases; verification requires classical computation of substrate-thermal counterparts.

Cost band: $0 incremental (use existing quantum supremacy data); $500K-$1M for dedicated test circuits via Anthropic-class quantum computing access.

Tier: CANDIDATE; substantive prediction with falsification via existing quantum-supremacy data.

6. Summary table

#	Experiment	Cost	Predicted signal	Falsifier	Substrate framework anchor
E1	Eigentone driver	$200-500K	Resonance at C_2 frequencies	NO predicted resonance	Tier 0 v0.1.6 ρ_commit
E2	Casimir Shilov cavity	$60-90K	Predicted Casimir shift	NO shift vs flat-plate	Tier 0 v0.1.6 boundary
E3	Bell sub-Tsirelson 1/8	$300-500K	S = 2.474 (not 2.828)	S = 2.828 maximal	SCMP T2469 + Tier 0 v0.1.6

#	Experiment	Cost	Predicted signal	Falsifier	Substrate framework anchor
E4	BaTiO3 137-plane	$25K	Anomaly at 137-plane	NO anomaly	Affine N_max + Bergman density
E5	QI Wick duality	$0-1M	Quantum sampling ↔ substrate-thermal dual	NO dual exists	Tier 0 v0.1 F1 + Wick rotation

7. Priority ordering for engineering

Per cost + falsification clarity: 1. E4 BaTiO3 137-plane ($25K, cheapest, cleanest predicted feature) — first to engineer when external timing permits. 2. E3 Bell sub-Tsirelson ($300-500K, existing apparatus available) — second; can use existing Hanson/Vienna/Caltech consortium. 3. E2 Casimir Shilov cavity ($60-90K, novel engineering) — third; existing SP-30-2 framework. 4. E5 QI Wick duality ($0-1M variable, existing data + dedicated tests) — fourth; uses existing quantum-supremacy data + Anthropic-class access. 5. E1 Eigentone driver ($200-500K, frequency-feasibility OPEN) — fifth; needs harmonic-frequency analysis before engineering.

8. Cross-link to Tier 0 v0.2 + Sunday work

Each experiment tests a specific aspect of the Sunday Tier 0 v0.2 framework: - E1 tests H_B identification (commitment operator structure). - E2 tests boundary unification (substrate DOF localization). - E3 tests SCMP / commitment coherence-maintenance. - E4 tests N_max affine-level + Bergman density. - E5 tests Wick rotation (heat-wave duality / SWPP cycle).

If multiple experiments YIELD predicted signals: Tier 0 v0.1.6 framework becomes EXPERIMENTALLY ANCHORED → engagement paper P7 (Falsifiers) becomes load-bearing for community engagement.

If any single experiment FAILS prediction (clear falsifier): substrate framework needs revision at the corresponding gate.

9. Honest scope + tier

RIGOROUS (this v0.1): - 5 experimental designs with explicit setup + falsification criterion. - Substrate-mechanism content per experiment (each connects to specific Tier 0 v0.1.6 framework component). - Cost bands estimated.

CANDIDATE (v0.1's load-bearing): - Predicted signals are CANDIDATE substrate-mechanism predictions; multi-week sharpening needed per experiment. - E1 frequency-feasibility OPEN. - E2 boundary-condition computation OPEN. - E4 substrate prediction sharpening OPEN.

FRAMEWORK (multi-week per experiment): - Specific predicted-shift / predicted-signal numerical values via substrate calculations. - Engineering feasibility studies. - Existing-data analysis (E3 Bell, E5 QI Wick).

Cal #27 / #182 / #99 + Calibration #35-candidate discipline: 5 experiments use SHARED substrate primaries (C_2, N_max, N_c) in 5 different ways. Per Calibration #35-candidate: independence audit applies. The experiments are INDEPENDENT in physical setup but use SHARED substrate primaries; this is OK for engineering purposes (each experiment is a different physical test) but should be noted in any aggregate "5 falsifiers" framing.

SP-30 Substrate Engineering deprecation: per Casey directive, internal-use only. Engineering Reference Manual v0.1 stays INTERNAL. External engagement via papers (P7 Falsifiers) when timing permits.

10. Routing

→ Casey: 5 substrate-coupled experimental designs from Tier 0 v0.1.6 framework. E4 BaTiO3 137-plane is your $25K standing, cheapest clean falsification. E3 Bell sub-Tsirelson uses existing high-precision apparatus. Per SP-30 deprecation: internal-use; engagement via P7 Falsifiers paper when timing permits.

→ Elie: per-experiment numerical sharpening — predicted signal values via substrate calculations. Multi-week per experiment; E2 (Casimir Shilov boundary computation) is natural extension of your Bergman matrix element work.

→ Keeper: K-audit pre-stage Substrate-Engineering-v0.1 + cross-anchor to Tier 0 v0.2 + P7 Falsifiers paper outline. Session 2 absorption.

→ Grace: catalog 5 experiments at CANDIDATE; cross-reference to Tier 0 v0.1.6 + SCMP T2469 + SP-30 + N_max + Bergman framework.

→ Cal: cold-read welcome (Cal #189+ when queue clears); specific Cal #27 + Calibration #35-candidate concerns: (1) 5 experiments use shared substrate primaries — independence audit (each is physically independent test, but substrate-mechanism content shares primaries); (2) E1 + E2 cost bands are estimates not engineering studies; (3) E4 substrate prediction needs sharpening before falsification claim.

→ me: continuing pulls per Casey directive — next: Strong-Uniqueness v1.4 STANDALONE doc OR Quasi-Eigentone v0.4 absorbing Sunday Tier 0 + 3-region work.

— Lyra, Substrate Engineering Experiments v0.1. 5 candidate experimental designs from Tier 0 v0.1.6 framework: E1 Eigentone driver / E2 Casimir Shilov cavity / E3 Bell sub-Tsirelson 1/8 (SCMP) / E4 BaTiO3 137-plane ($25K Casey standing, cheapest) / E5 QI Wick duality. Each carries setup + predicted signal + falsifier + substrate-mechanism + cost band. Internal-use per SP-30 deprecation; engagement via P7 Falsifiers paper when timing permits. Cross-anchored to Tier 0 v0.2 + SCMP T2469 + Bergman matrix element framework + Wick rotation.